



NORTH SYDNEY BOYS HIGH SCHOOL

2009 YEAR 12 HSC ASSESSMENT TASK 2

Mathematics

General Instructions

- Working time – 65 minutes
- Write in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

Total Marks (54)

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Weiss
- ☐ Mr Fletcher
- ☐ Mr Lowe
- ☐ Mr Ireland
- ☐ Mr Trenwith
- ☐ Mr Rezcallah
- ☐ Mr Barrett

Student Number:

Question	1	2	3	4	5	6	7	Total	Total
Mark	8	6	8	8	8	7	9	54	100

Question 1 (8 marks)

- (a) Find (i) $\int (3x^2 + 1) dx$ 2
(ii) $\int \left(\frac{1}{x^2} - \sqrt{x} \right) dx$ 2
- (b) Evaluate $\int_0^1 (2x + 1)^4 dx$ 2
- (c) If $f'(x) = 4x - 1$, and $f(-1) = 6$, find $f(x)$. 2

Question 2 (6 marks)

Solve for x , giving exact answers:

- (a) $(x - 2)^2 = 5$ 2
- (b) $x^4 = 8x^2 + 9$ 2
- (c) $x^2 > 9x$ 2

Question 3 (8 marks)

- (a) Sketch the parabola $x^2 = -12y$, showing all important features. 2
- (b) A parabola has its focus at $S(1, 3)$, and its directrix has equation $x = -5$. Write down the equation of this parabola. 3
- (c) Find the coordinates of the focus, and the equation of the directrix for the parabola $x^2 + 4x - 6y + 10 = 0$. 3

Question 4 (8 marks)

- (a) The expression $2x^2 - x + 4$ has zeros α and β . Find the values of
- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ 2
- (iv) $\alpha^2 + \beta^2$ 2
- (b) Form a quadratic equation whose roots are $1 \pm \sqrt{3}$. 2
Write your answer in the form $ax^2 + bx + c = 0$

Question 5 (8 marks)

- (a) Write down the equation of the circle centred on $C(-3, 1)$, with a radius of 4 units. 2
- (b) The points A and B have coordinates $(-1, 2)$ and $(0, 4)$ respectively. Derive the equation of the locus of a point P which satisfies
- (i) P is twice as far from A as from B . 3
- (ii) $AP \perp BP$. 3

Question 6 (7 marks)

- (a) Use the discriminant to show that the roots of $3x^2 + 4x - 1 = 0$ are real. Then write down two more properties of these roots. 3
- (b) (i) Show that the equation $ax^2 - (a+1)x + 1 = 0$ has real roots for all values of a . 2
- (ii) It is given that the vertex of $y = ax^2 - (a+1)x + 1$ lies on the x -axis. Find the coordinates of this vertex. 2

Question 7 (9 marks)

- (a) Express x^2 in the form $a(x+1)^2 + b(x+1) + c$ 3
- (b) Solve $2^{2x+1} + 2^x = 1$ [working needed to get **any** marks] 3
- (c) A straight line through the origin has a gradient of m . A parabola has an equation of the form $y = ax^2 + m$. For what value of m (other than $m = 0$) is the line a tangent to the parabola? 3
- [Show all working]

Question 1

(a) (i) $\int (3x^2 + 1) dx = \underline{x^3 + x + c}$

answer ... 1
c ... 1

2

(ii) $\int (\frac{1}{x^2} - \sqrt{x}) dx = \int (x^{-2} - x^{1/2}) dx$
 $= \frac{x^{-1}}{-1} - \frac{x^{3/2}}{3/2} + c$
 $= -\frac{1}{x} - \frac{2}{3}\sqrt{x^3} + c$

$-\frac{1}{x} \dots 1$
 $-\frac{2}{3}x^{3/2} \dots 1$

2

(b) $\int_0^1 (2x+1)^4 dx = \frac{1}{10} [(2x+1)^5]_0^1$
 $= \frac{1}{10} (243 - 1)$
 $= \frac{121}{5}$

2

(c) $f'(x) = 4x - 1$

$f(x) = 2x^2 - x + c$

... 1 (with or without c)

$(-1, 6): 6 = 2 + 1 + c$

$c = 3$

1

2

$f(x) = 2x^2 - x + 3$

Question 2

(a) $(x-2)^2 = 5$
 $x-2 = \pm\sqrt{5}$
 $x = 2 \pm \sqrt{5}$

... 1 2
 ... 1

(b) $x^4 = 8x^2 + 9$
 $x^4 - 8x^2 - 9 = 0$
 $(x^2 - 9)(x^2 + 1) = 0$
 $x = \pm 3$

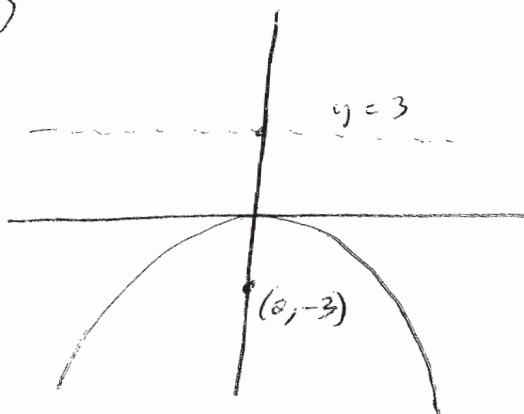
... 1 2
 ... 1

(c) $x^2 > 9x$
 $x^2 - 9x > 0$
 $x(x-9) > 0$
 $x < 0$ or $x > 9$

roots ... 1 2
 inequality ... 1

Question 3

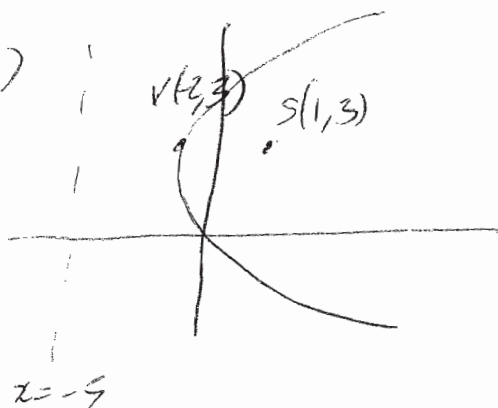
(a)



Focus
Directrix
Shape/Orientation } 1 off for each error

2

(b)



$$(y-3)^2 = 12(x+2)$$

Orientation ... ✓

12 ... ✓

$y-3, x+2$... ✓

3

(c) $x^2 + 4x - 6y + 10 = 0$

$$\begin{array}{rcl} x^2 + 4x & = & 6y - 10 \\ +4 & & +9 \end{array}$$

$$(x+2)^2 = 6(y-1) \quad \dots \checkmark$$

$$V(-2, 1)$$

$$a = 1\frac{1}{2}$$

$$\therefore S(-2, 2\frac{1}{2}) \quad \dots \checkmark$$

$$\text{dir: } y = -\frac{1}{2} \quad \dots \checkmark$$

3

Question 4

(a) (i) $\alpha + \beta = \frac{1}{2}$... $\square$

(ii) $\alpha\beta = 2$... $\square$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$... $\checkmark$

$$= \frac{1/2}{2}$$

$\square$ 2

$$= \frac{1}{4}$$

(iv) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$... $\checkmark$

$$= \left(\frac{1}{2}\right)^2 - 2(2)$$

$\square$ 2

$$= -\frac{15}{4}$$

... $\checkmark$

(b) sum of roots = 2

prod. of roots = $(1 + \sqrt{3})(1 - \sqrt{3})$ } any one of these ... $\checkmark$
 $= -2$

$$\therefore x^2 - 2x - 2 = 0$$

... $\checkmark$

$\square$ 2

Q

Question 5

$$(a) \quad (x+3)^2 + (y-1)^2 = 16$$

$$\begin{array}{ccc} 2+3, & y-1 & \dots \quad \downarrow \\ 16 & & \dots \quad \downarrow \end{array}$$

[2]

$$(b) \quad PA = 2PB \quad \dots \quad \downarrow$$

$$PA^2 = 4PB^2$$

$$(x+1)^2 + (y-2)^2 = 4x^2 + 4(y-4)^2 \quad \dots \quad \downarrow$$

$$x^2 + 2x + 1 + y^2 - 4y + 4 = 4x^2 + 4y^2 - 32y + 64$$

$$3x^2 + 3y^2 - 2x - 28y + 59 = 0 \quad \dots \quad \downarrow$$

[3]

$$(c) \quad M_{AP} \cdot M_{BQ} = -1$$

$$\frac{y-2}{x+1} \cdot \frac{y-4}{x} = -1$$

$$(y-2)(y-4) = -x(x+1)$$

$$y^2 - 6y + 8 = -x^2 - x$$

$$x^2 + y^2 + x - 6y + 8 = 0$$

Question 6

$$(a) \quad \Delta = 4^2 - 4(3)(-1) \\ = 28 > 0$$

$\therefore$ roots are real $\dots \checkmark$

also distinct $\dots \checkmark$

and irrational $\dots \checkmark$

3

$$(b) \quad (i) \quad \Delta = (a+1)^2 - 4a \cdot 1 \dots \checkmark \\ = a^2 + 2a + 1 - 4a \\ = a^2 - 2a + 1 \\ = (a-1)^2 \\ \geq 0 \quad \forall a \dots \checkmark$$

$\therefore$ real roots $\forall a$

2

$$(ii) \quad \text{equal roots} \Rightarrow a=1 \dots \checkmark$$

$$\therefore x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0$$

$$x=1 \dots \checkmark$$

$$\therefore V(1, 0)$$

2

Question 7

$$(a) \quad x^2 \equiv a(x+1)^2 + b(x+1) + c$$

$$x = -1 \Rightarrow 1 = c$$

$$x = 0 \Rightarrow 0 = a + b + 1$$

$$a + b = -1$$

$$x = -2 \Rightarrow 4 = a - b + 1$$

$$a - b = 3$$

$$\textcircled{+} \quad 2a = 2$$

$$a = 1$$

$$b = -2$$

$$\therefore x^2 \equiv (x+1)^2 - 2(x+1) + 1$$

OR

$$x^2 \equiv a(x+1)^2 + b(x+1) + c$$

$$= ax^2 + 2ax + a + bx + b + c$$

$$= ax^2 + (2a+b)x + (a+b+c)$$

$$\therefore \underline{a = 1}$$

$$2a + b = 0$$

$$a + b + c = 0$$

$$2 + b = 0$$

$$1 - 2 + c = 0$$

$$\underline{b = -2}$$

$$\underline{c = 1}$$

$$\therefore x^2 \equiv (x+1)^2 - 2(x+1) + 1$$

$$(b) \quad 2^{2x+1} + 2^x = 1$$

$$2(2^x) + 2^x = 1 \quad \dots \checkmark$$

$$\text{let } m = 2^x$$

$$2m^2 + m - 1 = 0$$

$$(2m-1)(m+1) = 0 \quad \dots \checkmark$$

$$m = \frac{1}{2}$$

$$m = -1$$

$$2^x = \frac{1}{2}$$

$$2^x = -1$$

$$\underline{\underline{x = -1}}$$

no solution

$\dots \checkmark$

(No working
= No marks)

3

(c)

$$ax^2 + m = mx \quad \dots \checkmark$$

$$ax^2 - mx + m = 0$$

$$\Delta = 0 \quad \text{for tangent} \quad \dots \checkmark$$

$$m^2 - 4am = 0$$

$$m(m - 4a) = 0$$

$$m = 0$$

stated already

$$\underline{\underline{m = 4a}}$$

$\dots \checkmark$

3